National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pr-University Examinations

Level: Bachelor | Semester: V Fall | Year : 2025
Program: BE Computer | F.M. : 100
Course: **Artificial Intelligence** | P.M. : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1 a) Can AI be conscious? Like can it have something like pain, awareness, emotions, selfhood? Justify. 7

b) Explain the four approaches to AI: Acting Humanly, Thinking Humanly, Thinking Rationally, Acting Rationally. 8

2 a. Describe PEAS representation with a real-world example. 8

b. Differentiate well-defined vs ill-defined problems with examples 7

3 a. Define CSP. Explain variables, domains, constraints, and backtracking search. Solve ABC + DEF = GHI. 9

b. Compare IDS with DFS and BFS. 6

4 a. Solve a search problem using A* algorithm with heuristic values: h(A)=10, h(B)=8, h(C)=5, h(D)=7, h(E)=3, h(F)=0 and Edges: A→B(6), A→C(3), B→D(2), B→E(4), C→E(7), C→F(10), D→F(1), E→F(5). With start node as A and goal node is F. 7

b. Convert the following statements into Propositional Logic and CNF then Prove the conclusion using Resolution. 8

i. If the system is secure, then data is protected.
ii. If data is protected and backups exist, then recovery is possible.
iii. If recovery is possible, the company continues operation.
iv. The system is secure.
v. Backups exist.

Prove: The company continues operation.

5 a. Perform K-means clustering (k=2) Points: (2,10), (2,5), (8,4), 8

(5,8), (7,5), (6,4). Take initial centroids as: (2,10), (5,8). What are applications of K means algorithm?

b. Explain Artificial Neural Network architecture and biological inspiration.

6 a. Explain Architecture of Expert System. 7

b. For given Fuzzy sets, A = {(1,0.2), (2,0.5), (3,0.9)} and B = {(1,0.7), (2,0.4), (3,0.3)}. Find: Union, Intersection, Complement 8

7 Write short notes on (any two) 10

a. Turing test VS Rational Agent
b. Rule based Systems
c. Backpropagation